Price Rigidity within Firm-to-firm Relationships

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Abstract

We study the determinants of price rigidity within firm-to-firm relationships. At the product level, we document significant heterogeneity in the frequency of price changes from a given supplier to different clients. Our results show that, for a given product, supplier-buyer characteristics drive an important fraction ($\approx$ 40%) of the heterogeneity in price adjustment among firms. We then show that suppliers with a higher market share in their clients' purchases are more likely to adjust prices in the face of oil price shocks. The effect is stronger for positive than negative shocks, highlighting the role of bilateral market power in price-setting behavior. We rationalize our results in a two-sector model with heterogeneous suppliers, menu costs, and non-constant demand elasticities.

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1 Introduction

Inflation dynamics and the real effects of monetary and fiscal policy hinge on the degree of price rigidity in the economy. The sluggishness with which prices respond to shocks depends on firm-level characteristics, such as the number of products sold (Midrigan, 2011), the degree of competitiveness in the market (Amiti et al, 2019; Mongey, 2023), and the sectoral composition of the economy (Pasten et al, 2023). Firms, however, are engaged in complex production chains, and their structure shapes the rigidity and the magnitude of price changes from one supplier to its different customers.

In this paper, we study the sources of price rigidity within firm-to-firm relationships. We particularly focus on the role that firm linkages and bilateral market shares play in shaping granular price adjustment decisions. For doing so, we use Chilean transaction-level micro data on prices at the seller-buyer-variety level. Our analysis proceeds in several steps. First, we document the aggregate evolution of price adjustment frequency, with particular attention to the shift observed during the COVID-19 pandemic. Next, we decompose the variance in price adjustment frequency across supplier, product, client, and supplier-client-specific effects, revealing that a substantial share of this variation is attributable to supplier-client characteristics.

We then conduct a regression analysis of price adjustment frequency on observable firm characteristics, including firm size, market share, and bilateral market importance. We find that, while firm size plays a role, the bilateral market share—the importance of a supplier in a buyer's total purchases—emerges as a key predictor of how frequently prices are adjusted, especially during the COVID-19 period, a period characterized by an endogenous increase in the frequency of price adjustments (Cavallo, Lippi, and Miyahara, 2024, e.g,). This finding suggests that price rigidity is not merely a product of firm-level attributes but is strongly influenced by the specific structure of inter-firm linkages.

While the frequency of price changes is important, it is ultimately the conditional response of prices to shocks that matters most for macroeconomic outcomes. Consistent with recent work by Cavallo, Lippi, and Miyahara (2024) and Blanco et al. (2024), our results indicate that the frequency of price adjustment shapes the pass-through of shocks. However, in contrast to previous studies, we show that an important driver of the frequency of price adjustment depends on supplier-client relationship characteristics. Specifically, we analyze the transmission of oil price shocks within our dataset. We find that suppliers are more likely to adjust the prices they charge to buyers for whom they represent a larger share of inputs, underscoring the importance of relationship-specific dependency in determining the extent of price pass-through.

In our final section, we discuss the implications for economic theory. We propose a simple model in the spirit of Gopinath and Itskhoki (2010) (i.e., menu costs and non-constant demand elasticity) but adapted to supplier-client relationships affected by oil price shocks. In the model, suppliers with higher market share in their buyer purchases of that input, face a more inelastic demand, which increases their probability of adjusting prices in the face of oil price shocks. Consistent with our evidence, these suppliers also change prices more, conditional on adjusting.

Our paper contributes to the literature studying sources of price setting using micro-data (e.g., Nakamura and Steinsson, 2008; Goldberg and Hellerstein, 2011; Bhattarai and Schoenle, 2014; Amiti, Itskhoki, and Konings, 2019; Hong et al., 2023). Our contribution is to provide a comprehensive description of the determinants of firm-to-firm price rigidity, which jointly considers supplier, client, and supplier-client characteristics. Our results complement the evidence documented in Goldberg and Hellerstein (2011) on the role of large firms' price setting. In addition, building on Pasten, Schoenle, and Weber (2020) we highlight the significant degree of heterogeneity in price rigidity among supplier-client industry pairs.

Our work also contributes to the literature on multi-sector models with sticky prices and input-output networks (e.g., Pasten, Schoenle, and Weber, 2020; La'O and Tahbaz-Salehi, 2022; Rubbo, 2023; Afrouzi and Bhattarai, 2023; Ghassibe, 2021). While these studies emphasize sectoral heterogeneity in price stickiness, a simple implication of our results is that, sectoral-level price stickiness ought to properly weight the fact that suppliers in a given sector can display different degree of price stickiness to firms in other sectors. The interaction between firm-to-firm level stickiness and networks can generate non-linear effects that aggregate in non-trivial ways.

More specifically, our findings challenge the traditional view that network structure and price rigidity can be considered in isolation. Instead, our results suggest that the rigidity of prices within a sector or firm is inherently linked to the structure of its network, particularly in the face of granular shocks–such as an energy transition shock–that disproportionately affect specific firms.

Our results relate to the recent literature that emphasizes the role of market structure in shaping the extensive margin of price adjustment to shocks and the degree of monetary non-neutrality, as in Mongey (2021). Our findings complement Mongey's by documenting that a two-sided market structure is important in shaping supplier-client price setting dynamics.[1]

Finally, our work contributes to the literature investigating the role of supplier-client relationships in shaping firms pass-through of shocks (e.g., Alviarez et al., 2023; Fontaine, Martin, and Mejean, 2020). Our key contribution is that we focus on the extensive margin of price adjustments in a context of menu costs to adjusting prices.

2 Data

Our data source consists of a comprehensive firm-to-firm transaction-level dataset, encompassing the universe of firms in Chile. Formal businesses in Chile must submit their

[1] Alviarez et al. (2023) document the importance of two-sided market power in the exporter-importer relationship. On the other hand, Burstein, Cravino, and Rojas (2024) use similar data to study the determinants of price dispersion, at a point in time, within a supplier to its different customers. The authors document that observables, such as product type and buyer-seller characteristics explain only a small fraction of the observed dispersion. However, the authors focus on the price level or magnitude of price change, while our focus is on the extensive margin of price setting.

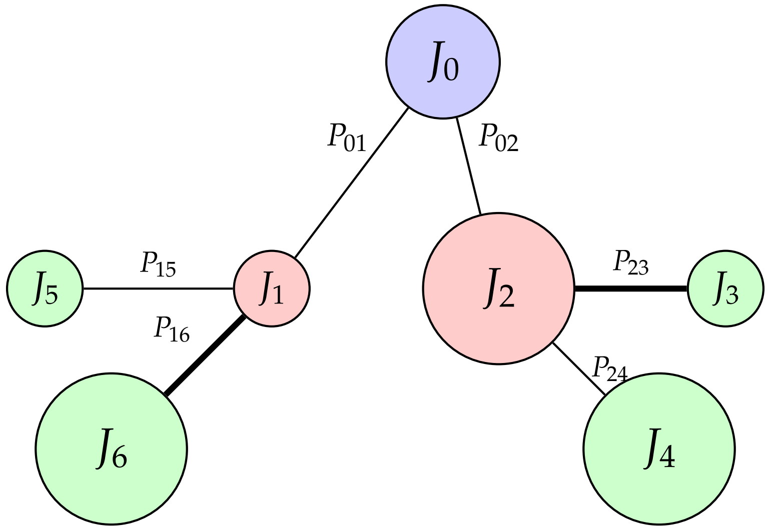
electronic receipts to the Chilean Internal Revenue Service (SII) compliance with regulatory mandates since 2018. The main variables we have access to are firms' unique tax identification numbers of both the supplier and the buyer, the date of the transaction, the total value of the transaction, the price per *unit*, the type of product involved in the transaction, and the location of the establishment.[2]

Figure 1 describes our data within firm-to-firm transactions. For simplicity, let us assume that each firm supplies one type of product. In the graph, we observe supplier J_0 selling its product to two different buyers, J_1 and J_2. The price charged to each buyer can differ and it is represented by P_{01} and P_{02}. The size of the node indicates the size of the firm and the thickness of the arrow represents the characteristics that are specific to the supplier-buyer pair (e.g., intensity, frequency, duration of the relationship). For example, supplier J_1 is a small supplier selling to a small buyer (J_5) and a large buyer J_6. The thicker arrow from J_1 to J_6 indicates a stronger relationship between supplier 1 and buyer 6, compared to supplier 1 and buyer 5.[3]

[2]To secure the privacy of workers and firms, the CBC mandates that the development, extraction, and publication of the results should not allow the identification, directly or indirectly, of natural or legal persons. All the analysis was implemented by the authors and did not involve nor compromise the IRS.

[3]Mongey (2021) emphasizes the role of firm-level market share in driving price adjustments. Alviarez et al. (2023) show how supplier and buyer market share within firm-to-firm relationships shapes price adjustments. Heise (2024) documents the importance of the age of the relationship between a supplier and a buyer in shaping shocks' transmission to prices.

Figure 1: Network graph of an oil supplier (J_0).



Note: network graph of a supplier J_0. J_1 and J_2 are customers of J_0, while J_3 and J_4 are customers of J_2, and J_5 and J_6 are customers of J_1.

Our dataset is enriched with balance sheet information for both sellers and buyers involved in the transactions. This supplementary data includes total sales, employment figures, industry classifications, and input purchases, among other variables.

Table 1 shows the descriptive statistics of annual sales, numbers of customers, and numbers of product subclasses sold. Subclasses are defined by the 173 aggregated product categories used by the National Institute of Statistics in Chile. The median firm in the sample displays annual sales of 250 CLP millions, 3.3 customers, and 1.3 product subclasses.

Table 1: Descriptive statistics (supplier characteristics)

	Mean	Std. Dev.	p10	p25	p50	p75	p90	Obs.
Average sales (CLP millions)	7,050	133,640	35.19	87.13	250.45	882.1	3,657	15,568
Number of customers	45.5	966.7	1.0	1.3	3.3	11.4	35.5	15,568
Number of products sold	2.4	3.2	1.0	1.0	1.3	2.5	4.8	15,568

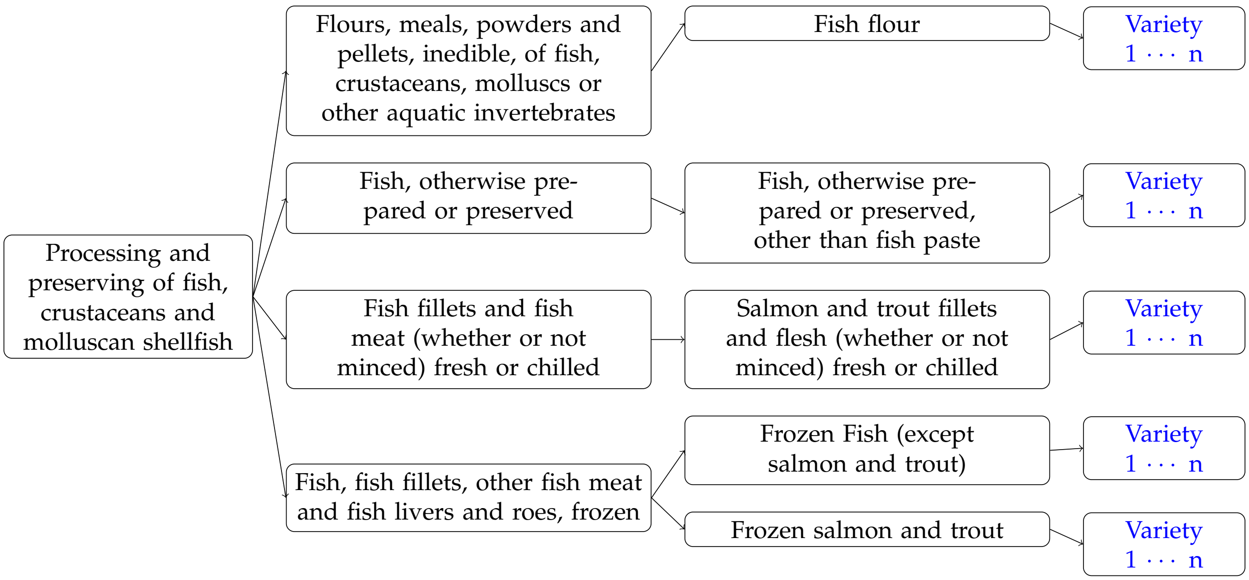
Consistent with the literature (e.g., Gabaix, 2011), the distribution of firm size is highly skewed. There is a long right tail with a handful of very large firms, implying an average sales that is almost thirty times larger than the median sales. A similar pattern is observed for the number of customers each supplier has. While the median firm has 3.3 customers, the average firm sells inputs to 45.5 customers. The average number of product subclasses sold by a supplier is 2.4, while the median is 1.3.

Here we describe the degree of disaggregation of transaction price data. One complication is that the reported information on the type of product or service traded among firms is self-reported and not always using standardized product definitions. For example, there might be two firms selling the same type of milk but one could report it as "two percent milk 1 liter" and the other as "2% milk 1lt". In this example, one would identify these as two different products. To overcome this limitation, we follow the methodology for data cleaning and processing outlined by Acevedo et al. (2023).

Acevedo et al. (2023) classify individual transactions from the firm-to-firm dataset into standardized product categories, such as the Individual Consumption Classification by Purpose (ICCP). This approach employs a supervised machine learning model on product descriptions and other observables, enabling the assignment of each transaction to a product category contained within the Consumer Price Index (CPI). Figure 2 provides an example of the level of granularity in our data. At the product class we have processing and preserving of fish, crustaceans and molluscan shellfish. These classes are further split into subclasses (e.g., fish, otherwise prepared or preserved), products (e.g., fish flour), and varieties (e.g., tuna fish flour).

To maintain a focus on significant and recurrent transactions, we have applied the following criteria to the dataset. Varieties included in our analysis are restricted to those associated with products in the official Consumer Price Index (CPI) and Producer Price Index (PPI) baskets. Moreover, each variety must be present in the dataset at least 24 times across any supplier to be considered in our analysis, allowing us to exclude sporadic or non-representative transactions. Price changes ($d \log P$) at the extreme percentiles (1 and 99) have been excluded to eliminate outliers and ensure a robust examination of price movements.

Figure 2: Example of granularity in price data: Classes, Subclasses, Products, and varieties



2.1 Validating price data with official data

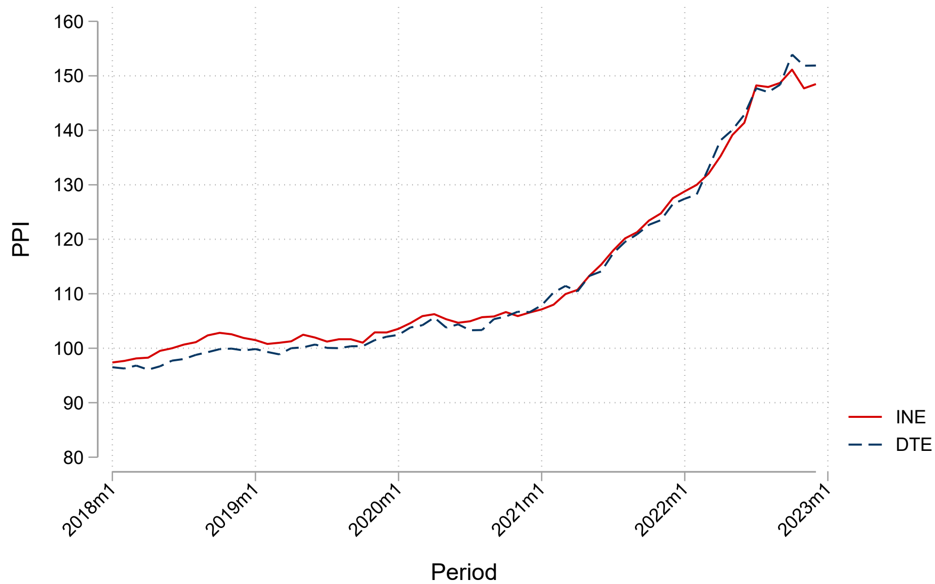
We now proceed to describe the validity and representativeness of our data. We compare the evolution of the official producer price index (PPI^{INE}) and the index we construct using our series of prices (PPI^c), which we define as

$$PPI_t^c = \sum_s \omega_s^{INE} P_{st},$$

where ω_s^{INE} is the PPI weight of subclass s in the official PPI by the INE, and P_{st} is the price of the subclass s at time t. We have access to daily transaction-level data which we aggregate to monthly prices following the methodology used by the National Institute of Statistics in Chile (INE). In particular, the subclass price is a geometric average of firm-level prices selling the subclass during a given month.

Figure 3 depicts the evolution of the official PPI index and the PPI we construct. We are able to successfully replicate the evolution of the official PPI. There are some small differences in levels but the correlation between the series is 0.99.

Figure 3: Producer Price Index (PPI): official vs constructed



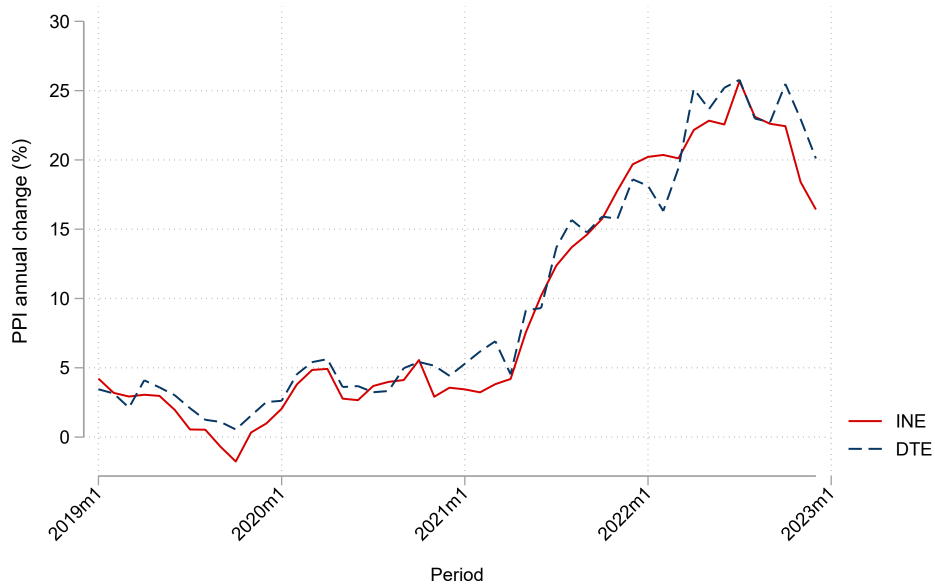
Note: The sum of weights is 0.97 of 100, and there are 165 of 173 products

The next Figure shows the implied annual percent change in prices (inflation). Our annual inflation measure is defined as

$$\pi_t^{PPI} = \sum_s \omega_s^{INE}(p_{st} - p_{st-1})$$

where p_{st} denotes the log of P_{st}. Figure 4 shows the evolution of PPI's official annual inflation and our proxy of it. We can trace down actual PPI inflation quite well. The correlation between the series is 0.98.

Figure 4: Annual PPI inflation: official vs constructed



Note: The sum of weights is 0.97 of 100, and there are 165 of 173 products

3 Facts on the Frequency and magnitude of price changes

In this section, we provide facts on firms' frequency price adjustment. We take advantage of the granularity of our data and present results at the supplier-client-variety level. Throughout the paper, we also provide facts at the supplier-client-subclass level. We aggregate slightly to show that our results are not driven by potentially mischaracterized varieties of a same product.

We define the frequency of price adjustment f_{ijv} as

$$f_{ijv} = \frac{\sum_{t=1}^{T_{ijv}} \mathbf{1}(|\Delta p_{ijvt}| > 0.005)}{T_{ijv}}, \tag{1}$$

in which $\mathbf{1}(|\Delta p_{ijvt}| > 0.005)$ is an indicator variable that takes the value of one if the price charged by supplier i to client j for variety v changed by more than 0.5%, and zero otherwise.[4] Here we have two options in defining Δp_{ijvt}. The first definition considers

[4] We use different thresholds and the results stay the same.

non-consecutive transactions, implying $\Delta p_{ijvt} = p_{ijv,t} - p_{ijv,t-\gamma_{ijt}}$, where γ_{ijvt} represents the time span between transactions. The second definition only considers the triplets ijv that display transactions in consecutive months. Therefore, in this case, we have $\Delta p_{ijvt} = p_{ijv,t} - p_{ijv,t-1}$. Throughout the paper, we focus on the first definition. In the Appendix, we use the second definition. All the main results stay the same.

In Table 2 we present the main descriptive statistics for the frequency of price change in our PPI data. The average price frequency is 0.34 and the median is 0.25. Note that the average frequency of price change at the supplier level is similar to that at the supplier-client-variety level.[5] As we will show next, these averages mask important heterogeneity in price adjustment at the supplier-client-variety level. The average frequency of price adjustment of 0.34 is somewhat larger than the average values in previous papers. However, as we show below, an important driver of this fact is the rise in the frequency of price changes during COVID-19.

Table 2: Frequency price change at supplier f_i and supplier-client-variety level f_{ijv}

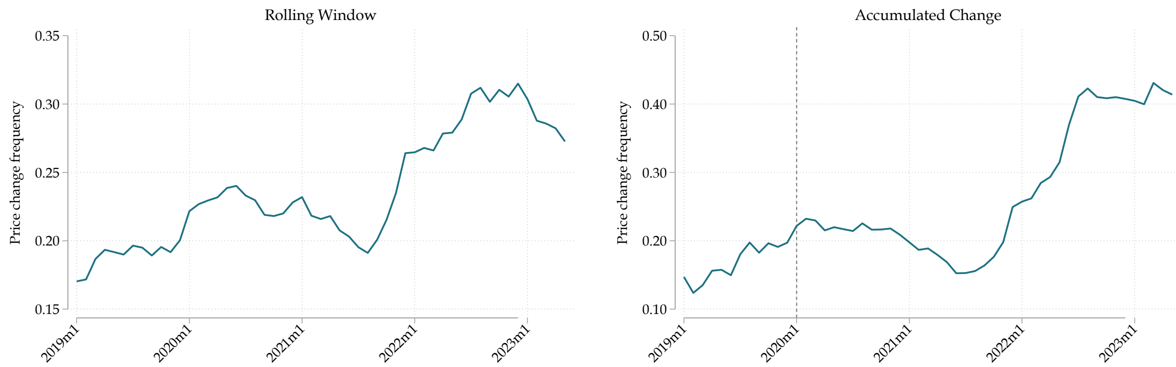
	Mean	SD	P10	P25	P50	P75	P90	N
f_i	0.34	0.28	0.05	0.13	0.25	0.50	0.84	15,568
dp_i	0.13	0.19	0.05	0.07	0.10	0.14	0.21	15,568
f_{ijv}	0.33	0.29	0.00	0.10	0.25	0.50	0.77	10,348,986
dp_{ijv}	0.31	0.26	0.00	0.12	0.25	0.45	0.71	8,014,515

Note: We measure the frequency price change f_{ijv} according to Equation (1). There are 15,568 suppliers in our dataset and 10,348,986 supplier-client-variety triplets.

Consistent with the findings in Blanco et al. (2024) and Cavallo, Lippi, and Miyahara (2024), the frequency of price adjustment significantly increased during COVID-19. In Figure 5 we observe that before COVID-19, prices adjusted 20% of the time. However, during COVID-19, the accumulated frequency of price changes (panel b), doubled to 40%.

[5] At the supplier level, we construct a simple supplier's price index averaging across varieties and buyers.

Figure 5: Absolute frequency change of prices



To assess the role of the extensive margin of price changes in inflation we perform the following decomposition, motivated by **?**. Let us define annual PPI inflation as follows:

$$\pi_t^{PPI} = \sum_s \omega_s (p_{st} - p_{st-1}),$$

$$\pi_t^{PPI} = \underbrace{\left(\sum_s \sum_t \omega_s I_{st} \right)}_{\text{Fraction of changing prices}} \underbrace{\left(\frac{\sum_s \sum_t \omega_s (p_{st} - p_{st-1})}{\sum_s \sum_t \omega_s I_{st}} \right)}_{\text{Average change, conditional on a change}},$$

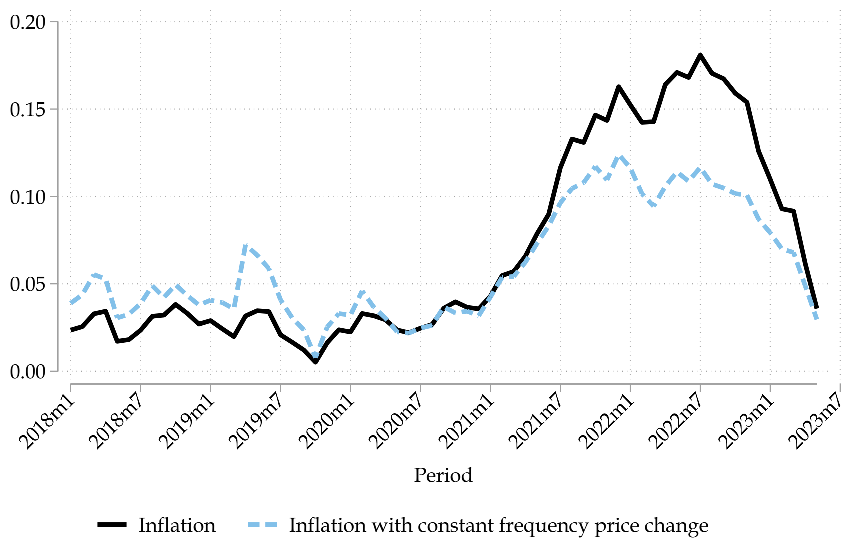
in which I_{st} is an indicator variable that takes the value of one when the price of product s changed more than a threshold ϵ ($abs(p_{st} - p_{st-1}) > \epsilon$), and zero otherwise. We use a threshold rather than zero as we construct p_{st} using geometric averages, meaning that if only one firm selling product s changes the price, I_{st} would equal one for all firms.

For the decomposition we follow Blanco et al. (2024). We compare the evolution of inflation against the inflation that would prevail if $\sum_s \sum_t \omega_s I_{st}$, the fraction of products changing prices stays constant and equal to its time average $\overline{\sum_s \sum_t \omega_s I_{st}}$. Specifically, we have

$$\overline{\pi}_t^{PPI} = \overline{\left(\sum_s \sum_t \omega_s I_{st} \right)} \left(\frac{\sum_s \sum_t \omega_s (p_{st} - p_{st-1})}{\sum_s \sum_t \omega_s I_{st}} \right),$$

We analyze annual inflation and use $\epsilon = 5\%$. Figure 6, shows that frequency price change played an important role in driving inflation during the 2021-2023 period. If firms maintained the historical average frequency of price change, peak inflation during 2023 would have been 40% lower. These results are consistent with those in González and Rojas (2023), who using a slightly different approach, show that monthly inflation is largely driven by the extensive margin of prices.[6]

Figure 6: Decomposition of annual PPI inflation: the role of the extensive margin of price adjustment



Note: Frequency of price change is defined for annual index changes greater than 5%.

4 Price facts within supplier-client relationships

We intend to understand the role of supplier-client relationships in shaping the frequency of price adjustments. Fontaine, Martin, and Mejean (2020), Dhyne, Kikkawa, and Magerman (2022), and Alviarez et al. (2023) have emphasized the importance of match-specific characteristics in shaping the heterogeneity in the level of prices. In the presence of menu costs, market structure can also affect the frequency at which prices adjust (e.g., Gopinath and Itskhoki, 2010; Mongey, 2021). Hence, we study how price rigidity at the relationship level is determined by bilateral market shares, among other supplier-client characteristics.

[6]In their analysis, the authors consider a price change only when the specific variety, within the product subclass, changed more than 10%, monthly.

We measure the suppliers and buyers market shares following Alviarez et al. (2023). In particular, the supplier's market share is defined as

$$s_{ijvy} = \frac{p_{ijvt}q_{ijvt}}{\sum_{k \in Z_j} p_{kjsy}q_{kjsy}},$$

where $p_{ijvy}q_{ijvy}$ represents the sales of product-variety v from seller i to buyer j during year y. The denominator $\sum_{k \in Z_j} p_{kjst}q_{kjst}$ represents the purchases of any input within the subclass s made by buyer j from all possible suppliers of firm j labelled as Z_j.[7] We then measure the market share of a given buyer j in the sales of a supplier i (x_{ijvy}) as follows

$$x_{ijvy} = \frac{q_{ijvy}}{\sum_{k \in Z_i} q_{iksy}}.$$

In Appendix 7.4 we show the descriptive statistics of this new measure, in particular the distribution of s_{ijvt} and x_{ijst}.

Building on our granular data of the ij match characteristics, we also observe the number (#) of transactions observed between a given seller-buyer and the length of the relationship between the pair.[8]

Given the unexplored features of our measures on sellers and buyers market shares, we study their consequences on price-adjustment first at the cross-sectional level and then by taking into account the full panel dimension.

4.1 Cross Sectional Evidence

We begin our empirical analysis by estimating a series of cross-sectional regressions that explore the average determinants of price rigidity across supplier-buyer-varieties triplets.

[7] Following Figure 2, we interpret s_{ijvy} as the total purchases of buyer j from seller i of a specific product-variety of v, e.g. $v =$"fish flour" among *the total* purchases of j (from any supplier) of products within the subclass s, e.g. $s =$ "Flours, meals, powders and pellets within year y. An alternative, more aggregated measure, uses subclasses in the numerator and total intermediate purchases in the denominator.

[8] Heise (2024) documents a positive relationship between the price pass-through (intensive margin) in exporter-importer relationships and the age of that relationship. The author explains this pattern in a model displaying customer relationships in firm-to-firm relationships.

This approach provides a static assessment of the main determinants of price adjustments and how they depend on the structure of relationships within the firm's network. This exercise provides an initial, time-invariant description of the main features of firm-to-firm connections and their correlation with pricing outcomes.

Specifically, we solve the following equation:

$$y_{ijv} = \alpha + \beta \mathbf{X} + FE_s + FE_{Ind_{I_i}} + FE_{Ind_{I_j}} + +\varepsilon_{ijv} \tag{2}$$

Where we allow the dependent variable to change to measure different angles of individual prices at their flexibility at the ijv level. In particular, the dependent variable y_{ijv} includes the log level of prices $\ln(p_{ijv})$, a dummy equal to one if $\Delta p_{ijv} \neq 0$, and a dummy if prices changed upwards or downwards $\Delta p_{ijv} > 0$ and $\Delta p_{ijv} < 0$, respectively. Hence, these last three expressions aim to characterize the extensive margin of price changes. For the intensive margin, we rely on the magnitude of price revisions conditional on adjustment $|\Delta p_{ijv}|\Delta p_{ijv} \neq 0|$ as the dependent variable.

The specification (2) includes fixed effects at the sellerâs subclass level (s) and both seller (I_i) and buyer industry (I_j) levels. Our controls in X include the bilateral market share variables (s_{ijv} and x_{ijv}), the number of transactions between ij, the length of the relationship, and the number of subclasses of products sold by the supplier. Importantly, we also include a relevant network-based regressor: the input price adjustment frequency or magnitude, computed as a weighted average of upstream pricing behavior for each seller s.Tables 4 and 3 show the results for the extensive and intensive margins.

Table 3: Cross-Section Evidence - Extensive Margin

	(1) $\Delta p_{ijv} \neq 0$	(2) $\Delta p_{ijv} > 0$	(3) $\Delta p_{ijv} < 0$
s_{ijs}	0.018***	0.013***	0.005***
	(0.000)	(0.000)	(0.000)
x_{ijs}	0.056***	0.003*	0.053***
	(0.002)	(0.002)	(0.001)
# Transactions	-0.171***	-0.153***	-0.018***
	(0.000)	(0.000)	(0.000)
Duration	0.159***	0.144***	0.015***
	(0.000)	(0.000)	(0.000)
3-7 subclasses	-0.012***	0.005***	-0.017***
	(0.000)	(0.000)	(0.000)
7+ subclasses	0.030***	0.026***	0.004***
	(0.000)	(0.000)	(0.000)
Input Freq. Change	0.255***	0.124***	0.132***
	(0.001)	(0.001)	(0.000)
Constant	0.109***	0.105***	0.004***
	(0.001)	(0.000)	(0.000)
N	10,301,452	10,301,452	10,301,452
Adjusted R2	0.425	0.350	0.279
Within R2	0.182	0.190	0.029

Table 4: Cross-Section Evidence - Intensive Margin

	Ln(p_{ijv})	$\lvert\Delta p_{ijv}\vert\Delta p_{ijv} \neq 0\rvert$	$\lvert\Delta p_{ijv}\vert\Delta p_{ijv} > 0\rvert$	$\lvert\Delta p_{ijv}\vert\Delta p_{ijv} < 0\rvert$
s_{ijs}	0.056***	0.010***	0.009***	0.026***
	(0.002)	(0.000)	(0.000)	(0.001)
x_{ijs}	-0.612***	0.071***	0.066***	0.061***
	(0.013)	(0.003)	(0.003)	(0.006)
# Transactions	0.030***	-0.059***	-0.060***	-0.068***
	(0.001)	(0.000)	(0.000)	(0.000)
Duration	0.015***	0.046***	0.050***	0.050***
	(0.001)	(0.000)	(0.000)	(0.000)
3-7 subclasses	0.102***	-0.015***	-0.013***	-0.011***
	(0.003)	(0.000)	(0.000)	(0.001)
7+ subclasses	0.041***	-0.010***	-0.010***	0.003***
	(0.003)	(0.000)	(0.000)	(0.001)
Input Mag. Change	0.036**	0.024***	0.023***	-0.055***
	(0.018)	(0.003)	(0.003)	(0.008)
Constant	7.462***	0.117***	0.106***	0.127***
	(0.003)	(0.001)	(0.000)	(0.001)
N	10,301,452	8,268,397	7,954,598	3,430,054
Adjusted R2	0.446	0.069	0.078	0.075
Within R2	0.002	0.029	0.034	0.015

An important result emerges from both extensive and intensive margin specifications concerning the role of input-side dynamics. As noted, the frequency and magnitude of price changes in a supplier's upstream inputs are strongly associated with their own price adjustment behavior. In particular, a (hypothetical) shift from no input changes to constant changes over the sample period, i.e., Input Freq. Change $= 1$, increases the supplier's own probability of revising prices by approximately 26%. Similarly, a 1% increase in upstream prices is associated with a 2.4% increase in the supplier's output prices. Such effects can be observed in columns 1 and 2 of Table 3 and 4, respectively. This evidence brings two main implications. First, it provides direct empirical evidence of network propagation in pricing behavior: firms more exposed to upstream input price volatility are themselves more flexible in their pricing. Second, it highlights that such cost pass-through is, on average incomplete, consistent with the presence of frictions or other preferences-related factors that attenuate the full transmission of costs along the supply chain. In Section 5, we revisit this issue using local projections to estimate dynamic pass-through responses in the network.

We next examine the role of bilateral market power. Larger supplier market shares (s_{ijv}) are associated with higher prices and more frequent price adjustments. If a seller accounts for the entire share of a buyer's purchases in a given subclass (i.e., $s_{ijv} = 1$), the probability of an upward price revision increases by 1.3 percentage points, while that of a downward revision rises by 0.5 points. These magnitudes are meaningful: the median frequency of price changes across ijv triplets is 0.25 (see Table 2) in our sample, which even includes the inflationary surge during the pandemic. Interestingly, the increase in downward adjustments may reflect the fact that our sample spans both inflationary and disinflationary periods—an interpretation we revisit in the panel regressions below.

In contrast, larger buyer market shares (x_{ijv}) are associated with lower prices and a higher frequency of downward price revisions. These results align with the intuitive notion that bigger and powerful buyers can bargain for lower prices, which are therefore more likely to decrease. Still, the sign of some effects again suggests that unobserved heterogeneity may be affecting some of the estimates, which motivates the panel approach in the next section.

Finally, other relationship characteristics also matter. More frequent transactions are associated with less frequent price revisions, consistent with the presence of explicit or implicit contracts that prevent continuous renegotiation. Meanwhile, longer relationships are correlated with higher prices, plausibly reflecting the fact that longer-standing links have had more time to incorporate cumulative price revisions.

4.2 Panel Data Evidence

While the cross-sectional regressions offer insight into the average structural determinants of price rigidity, they cannot account for relevant unobserved heterogeneity across firms, products, and time. We therefore turn to the following panel estimation:

$$y_{ijvt} = \alpha + \beta X + FE_i + FE_j + FE_s + FE_{Ind_{I_i}} + FE_{Ind_{I_j}} + FE_t + \varepsilon_{ijvt} \tag{3}$$

Where again the dependent variables are the same as in the cross-section, $y_{ijv} = \{Ln(P), \Delta p_{ijv} \neq 0, \Delta p_{ijv} > 0, \Delta p_{ijv} < 0, |\Delta p_{ijv}|\Delta p_{ijv} \neq 0|$ as well as the controls in the vector X. However, as we add dynamics through the panel specification, we also add other controls such as the gap to competitors and the age of a price. The inclusion of the price gap and price age variables allows us to interpret pricing behavior as a function of state variables. The former reflects deviations from a benchmark price level (e.g., the average across competitors), while the latter captures how "stale" the current price is. These indicators are strongly associated with adjustment probabilities, highlighting the role of state-dependent pricing, Campbell and Eden (2014); Karadi, Schoenle, and Wursten (2024).

With respect to the seller and buyers market share, given their high persistence, we measure them at the yearly level y, s_{ijvy}, and x_{ijvy}. We saturate regression 3 with a broad set of fixed effects: seller (i), buyer (j), subclass (sub), industry (for both sides) $I_{Ind_{I_i}}$, $I_{Ind_{I_j}}$, and time t (including month and period FEs). These controls aim to account for time-invariant firm characteristics while allowing us to isolate within-relationship variation over time. The results for the extensive and intensive margins are presented in Tables 5 and 6.

Relative to the cross-sectional setting, this approach emphasizes the role of within-pair dynamics while controlling for persistent unobserved heterogeneity. From Table 5, we confirm that both the gap to competitors and the age price matter for triggering price

Table 5: Panel Estimation - Extensive Margin

	(1) $\Delta p_{ijv} > 0$	(2) $\Delta p_{ijv} > 0$	(3) $\Delta p_{ijv} < 0$	(4) $\Delta p_{ijv} < 0$
Price gap (t-1)	-0.011***	-0.012***	0.000***	0.000**
	(0.000)	(0.000)	(0.000)	(0.000)
Price Age	0.025***	0.013***	-0.015***	-0.018***
	(0.000)	(0.000)	(0.000)	(0.000)
s_{ijsy}		0.017***		0.007***
		(0.000)		(0.000)
x_{ijvy}		-0.007***		0.006***
		(0.001)		(0.001)
# Transactions		-0.128***		-0.017***
		(0.000)		(0.000)
Relationship Duration		0.063***		0.013***
		(0.000)		(0.000)
Constant	0.202***	0.281***	0.090***	0.089***
	(0.000)	(0.000)	(0.000)	(0.000)
N	130,338,866	130,338,866	130,338,866	130,338,866
Adjusted R2	0.121	0.141	0.125	0.126
Within R2	0.003	0.026	0.003	0.004

adjustments. In particular, in a context of positive trend inflation as in our data, prices that have remained constant for several periods are more likely to be revised upwards than downwards, Ball and Mankiw (1994); Karadi and Reiff (2019). The bilateral market power continues to play a statistically and economically significant role. Supplier market share (s_{ijvy}) remains positively associated with both price levels and the frequency of upward adjustments. As observed in the second column of Table 6, the overall price p_{ijvt} is approximately 30% higher if seller i becomes the only supplier of variety v for firm j. Likewise, the probability of a price increase rises by 1.3% in this scenario. While there is still evidence that a higher s_{ijvy} is associated with a higher probability of negative pricing

Table 6: Panel Estimation - Intensive Margin

	(1) Ln(P)	(2) Ln(P)	(3) $\lvert\Delta p_{ijv}\lvert\Delta p_{ijv} \neq 0\rvert$	(4) $\lvert\Delta p_{ijv}\lvert\Delta p_{ijv} \neq 0\rvert$	(5) $\lvert\Delta p_{ijv}\lvert\Delta p_{ijv} > 0\rvert$	(6) $\lvert\Delta p_{ijv}\lvert\Delta p_{ijv} > 0\rvert$	(7) $\lvert\Delta p_{ijv}\lvert\Delta p_{ijv} < 0\rvert$	(8) $\lvert\Delta p_{ijv}\lvert\Delta p_{ijv} < 0\rvert$
Price gap (t-1)	0.028***	0.025***	0.004***	0.004***	0.003***	0.003***	0.002***	0.002***
	(0.001)	(0.001)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Price Age	-0.039***	-0.038***	0.027***	0.016***	0.027***	0.016***	0.029***	0.020***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
s_{ijsy}		0.323***		0.001***		0.001***		0.001
		(0.001)		(0.000)		(0.000)		(0.000)
x_{ijvy}		-1.019***		-0.018***		-0.023***		-0.007***
		(0.003)		(0.001)		(0.001)		(0.002)
# Transactions		-0.041***		-0.036***		-0.036***		-0.034***
		(0.000)		(0.000)		(0.000)		(0.000)
Relationship Duration		0.008***		0.014***		0.013***		0.011***
		(0.000)		(0.000)		(0.000)		(0.000)
Constant	7.671***	7.683***	0.055***	0.092***	0.051***	0.088***	0.066***	0.102***
	(0.000)	(0.001)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
N	130,338,866	130,338,866	38,779,375	38,779,375	29,636,020	29,636,020	9,115,784	9,115,784
Adjusted R2	0.706	0.708	0.117	0.128	0.135	0.148	0.200	0.204
Within R2	0.002	0.010	0.018	0.031	0.024	0.039	0.012	0.018

revisions too, from the last column of Table 6, we notice that the magnitude of revisions is not statistically significant.

Buyer market power (x_{ijvy}) is negatively associated with prices and the likelihood of positive adjustment. In fact, a higher buyer market power increases the probability of negative price revisions and of a bigger magnitude. We interpret the coefficient x_{ijvy} of column (6) in Table 6, as evidence that, every time a price increased, buyers with higher market power experience smaller increases compared to others.

Similar to the cross-section results, the number of transactions strongly reduces the probability of upwards revisions, where, even when prices increase, the revision magnitude is less the higher the number of transactions. Likewise, the implications of the relationship length are also robust relative to the previous results.

Taking stock, besides the unobserved patterns at the firm and time levels, the evidence suggests that specific match characteristics between sellers and buyers matter for price-adjustment decisions. Building on this fact, and leveraging the network structure, we aim to further characterize the dynamics of pass-through at different network positions while conditioning on ij matching characteristics.

5 Price adjustments to oil price shocks

So far, we have provided evidence of unconditional price changes and how these changes relate to firm-to-firm relationships. In this Section, we study the importance of the nature of firm-to-firm relationships in the transmission of a cost-push shock. In particular, we focus on oil price shocks along the production network in terms of the extensive margin and the intensive margin of price adjustment.[9]

In Appendix 7.3, we describe the main stylized facts that we use to study the conditional price reaction to Oil shocks. To gain intuition and to further explore the richness of our data, in all the following exercises, we focus on suppliers in node J_0, J_1, and J_2 of Figure 1. Node J_0 represents the main Oil Supplier company in the Chilean economy. The nodes underneath (J_1 and J_2 in the example of Figure 1) are different producers that directly re-sell a large proportion of crude oil to their customers. In particular, we classify suppliers at these nodes if their share of total sales of Oil relative to all sales is bigger than 0.5.[10]

Appendix 7.3 describes the two measures of seller market share and buyer market share at these levels. In particular, we split between the ij and the ijs bilateral market share. In the former, and in line with the definitions discussed in Section 4, we compute s_{ij} as the ratio of all sales from i to j relative to all purchases of j, while for s_{ijs} the numerator accounts for sales from i to j of products within the subclass s. As discussed, in this case, the subclass category corresponds to "Refined Oil Products". Similarly, and again in line with Section 4, x_{ij} and x_{ijs} are constructed as the ratio of total quantities offered from i to j relative to firm i's total production. According to the stylized facts, the seller's bilateral market share is sizeable, accounting for an average of 0.66 and 0.62 for s_{ij} and s_{ijs},

[9] Lafrogne-Joussier, Martin, and Méjean (2023) also studies the pass-through of oil price shocks, focusing on the intensive margin of adjustment at the firm level. The authors show that the pass-through is larger for larger firms, indicating oligopoly power in pricing decisions as in Atkeson and Burstein (2008).

[10] As expected, a typical firm at this node is a gasoline station, transport companies, or a manufacturing firm that redistributes oil to third parties. Recall that the different sizes of these two nodes are only for expository reasons, aiming to resemble different sellers' characteristics, such as sizes, sales, or number of employees.

respectively. On the contrary, buyers' market share is often small, with an average of 0.045 and 0.06 for x_{ij} and x_{ijs} in this case. As expected given the position in the network, once we condition on a ij match, most of the transactions of an Oil-related specific item tend to pervasively come from the same seller i. On the contrary, among all the i production of Oil products, less than 5% are sold to firm j.

5.1 The extensive margin adjustment

For the extensive margin of price adjustment, we follow Karadi, Schoenle, and Wursten (2024) and estimate the following regression

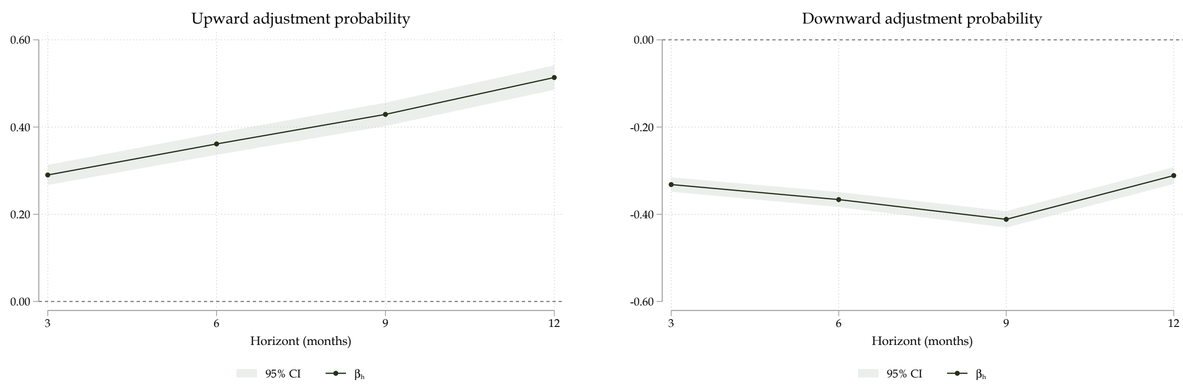
$$I^{+,-}_{ijv,t+h} = \alpha_i + \alpha_j + \alpha_s + \beta_h \Delta \log \hat{P}^{oil}_{it} + \phi_h(\Delta \log \hat{P}^{oil}_{it} \times Z_{ij}) + \gamma_h X_{ij} + \psi w_t + \epsilon_{ijs}, \quad (4)$$

In the equation, $I^{+,-}_{ijv,t+h}$ is a dummy variable that takes the value of one if the log change of prices from period $t+h$ compared to period $t-1$ for supplier i, selling a variety v to customer j, changed. The variable $\Delta \log \hat{P}^{oil}_{it}$ is the change in the log of oil price that is relevant for supplier i. This is, we measure the exposure of supplier i to oil as an input using our detailed data.

We instrument the change in oil prices using the oil supply shock from Baumeister and Hamilton (2019). Given the richness of our data, we interact the (instrumented) change in Oil price with Z_{ij}, which can represent i, j, or ij specific (time-invariant) characteristics. For what follows, we focus particularly on both the seller and buyer market share (but we could explore other ij specific match characteristics, as duration of the relationship or the number of transactions). We also add an additional set of controls X_{ij} for each of the specifications.

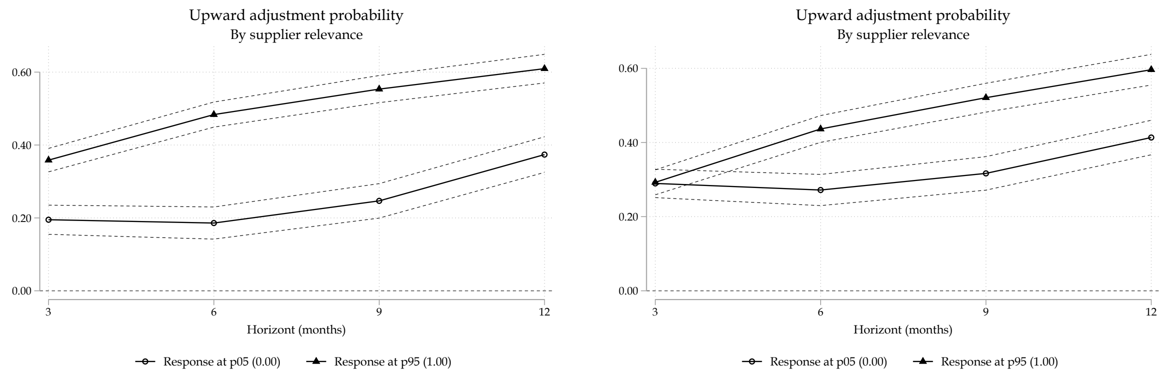
Figure 7 depicts the average probability of price adjustment after 3, 6, 9, and 12 months after the oil shock. In the estimation, we focus on sellers at the J_1 and J_2 nodes, and initially, we do not include the interaction term ϕ_h in 4. The left (right) panel plots the probability of an upward (downward) price revision after an oil price increases at the different horizons. The average probability of increasing prices after a 1% increase in oil prices is 30% after three months, and it increases up to 50% after 12 months. On the other hand, the probability of reducing prices is similar 3 months after the change. Still, unlike the upward adjustment probability, the downward adjustment probability stays around 30% even after 12 months.

Figure 7: Probability of price adjustment to a 1% increase in oil prices



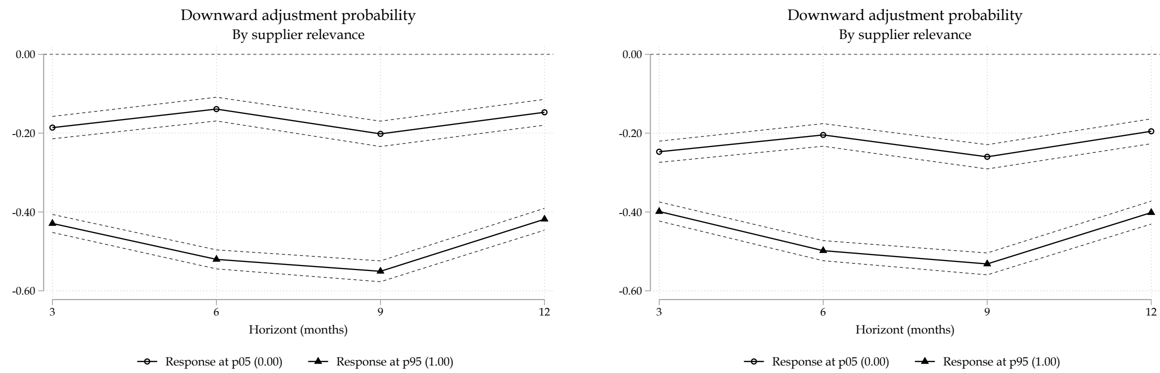
Building on this evidence, in Figure 8, we analyze the influence of supplier market shares on upward price adjustments. For this, we estimate (4) adding the interaction term, where Z_{ij} could be either the ij bilateral market share s_{ij} (left panel) or the subclass-specific market share s_{ijs} (right panel). In either case, we study the potential heterogeneous response conditioning on the market-share variable being at the 5th and 95th percentile values.

Figure 8: Upward Adjustment probability by Supplier Market Share



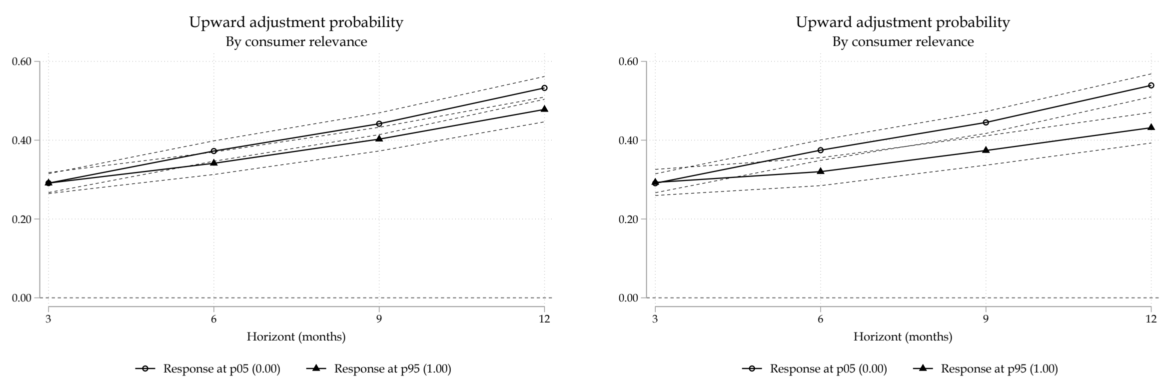
According to the results, suppliers with larger bilateral market shares (s_{ij} or s_{ijs}) are significantly more likely to increase prices to their customers after a change in Oil prices. Specifically, suppliers in the 95th percentile of the ij bilateral market share distribution have a three-month probability of raising prices that is twice as high—40% compared to 20%—as suppliers in the 5th percentile. This difference remains significant over a 12-month horizon, with dominant suppliers exhibiting a 60% probability of upward price adjustments, compared to a less than 40% probability for atomistic suppliers. Focusing on item-specific market shares s_{ijs}, the results are fairly similar except for the immediate effect 3 months after the price change, where the response is similar. Turning to the probability of downward adjustment, the results are similar, with a very clear different dynamics for sellers with a small relative to a large market share. This is shown in Figure 9.

Figure 9: Negative Price Change by Supplier Market Share



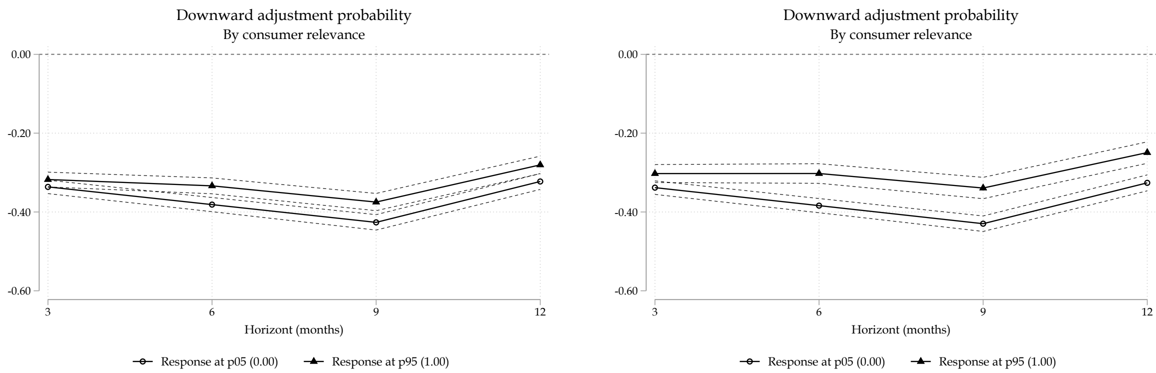
Up to this point, it is clear that the effect of seller-market share is non-negligible in explaining the extensive margin of price adjustment. We now study whether the ij consumer market share x_{ij} or its subclass-specific counterpart x_{ijs} matters for the price adjustment probability. The results for upward and downward adjustments using these two measures are shown in Figures 10 and 11. As before, left panels assume $Z_{ij} = x_{ij}$ while right panels set $Z_{ij} = x_{ijs}$.

Figure 10: Positive Price Change by Consumer Market Share



Distinct from the effects of seller market share, in this case, there are no apparent statistical differences for either upward or downward adjustments when we condition on small or large customer market shares. Thus, while it seems that bilateral market share matters, the heterogeneous effect of price adjustments is particularly stringent when we focus on sellers that exert more market power on their clients.

Figure 11: Negative Price Change by Consumer Market Share



5.2 The intensive margin adjustment

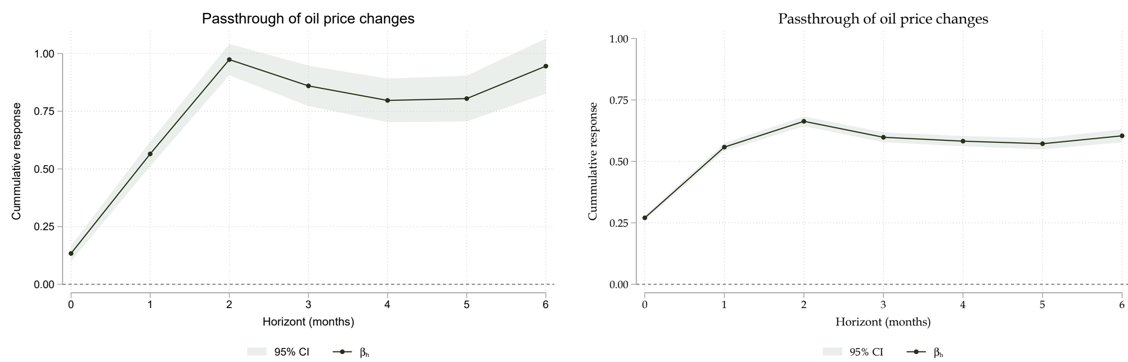
Building on the previous evidence, we now turn to studying the intensive margin of price adjustments. We follow Jordà (2005) and estimate the following Local Projection-IV regression

$$\pi^{K,ijv}_{t-1,t+h} = \alpha + \beta^K_h(\Delta \log \hat{P}^{oil}_{it}) + \phi^K_h(\Delta \log \hat{P}^{oil}_{it} \cdot Z_{ij}) + \sum_{j=1}^{12} \delta^j \pi_{t-j} + \sum_{j=1}^{12} \gamma^j \Delta P_{c,t-j} + \psi X_t + \varepsilon_t, \tag{5}$$

The dependent variable in equation (5) $\pi^{ijs}_{t-1,t+h}$, is the log change of prices from period $t+h$ compared to period $t-1$ for supplier i, selling product within subclass s to customer j. As before, $\log \hat{P}^{oil}_{it}$ is the instrumented log price change and Z_{ij} our market share controls. Following the same strategy as before, initially, we estimate (5) without the interaction term to study how the Oil price adjustment passes through to micro prices, β^K_h.

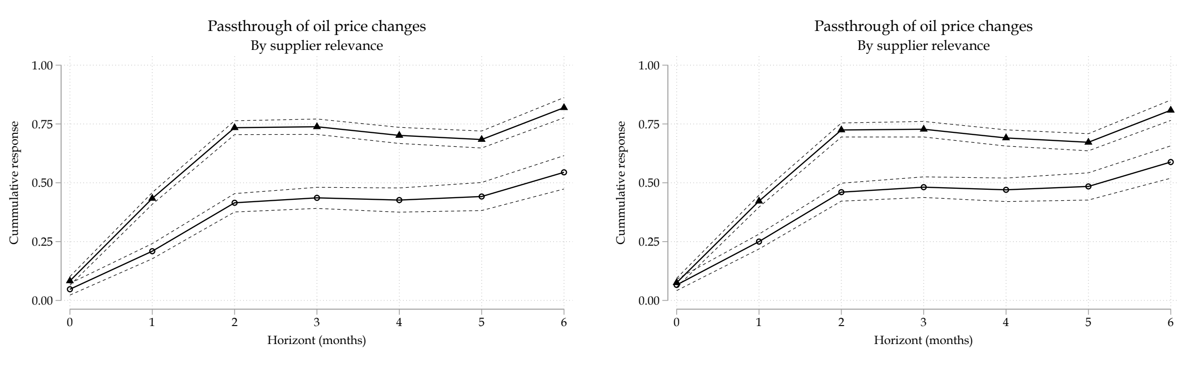
In Figure 12, we plot the intensive margin response of prices at different nodes in the network. The left panel focuses on the pass-through of the main oil Supplier company in the Chilean economy (which is by definition at node J_0 in Figure 1). The evidence supports the presence of a full pass-through after two months. The right panel shows the pass-through of firms that are the largest sellers of oil in the economy (again, the ones in nodes J_1 in Figure 1). In this case, the initial pass-through is 0.25, reaching almost 0.75 two months after the increase in Oil prices. Afterwards, the pass-through stays relatively constant for the upcoming months, pointing to an incomplete pass-through at this point in the network.

Figure 12: Price response of firms in oil extraction at initial node (J_0) and refinery industries (J_1)



Now, we incorporate the interaction term in (5). In this case, we are interested in the pass-through effect of oil prices to micro prices $\frac{\partial \pi^{K,ijv}_{t-1,t+h}}{\partial \Delta \log \hat{P}^{oil}_{it}} = \beta^K_h + \gamma^K_h Z_{ij}$, conditioning on different values of Z_{ij} across the market share's percentiles. Figure 13 presents the marginal effect using the two measures s_{ij} (left panel) and s_{ijs} (right panel) at node J_1.

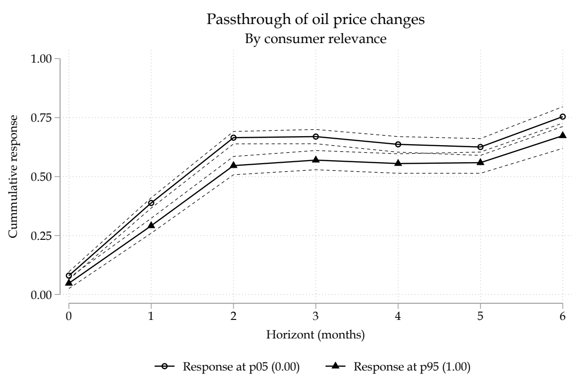
Figure 13: Intensive Margin by Supplier Market Share

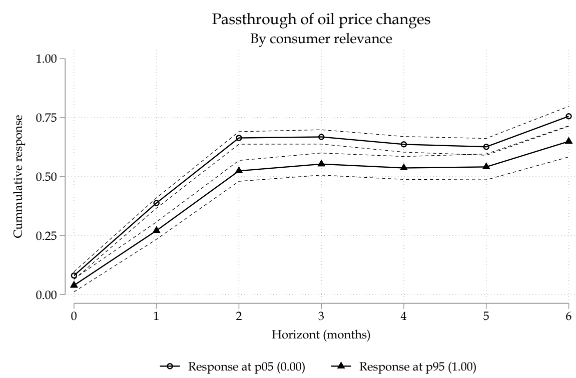


Consistent with the extensive margin, some clear differences arise. Suppliers with higher bilateral market share pass-through their higher costs more aggressively than firms with a small market share. In fact, two months after the oil price change, sellers with higher market power have already passed through 0.75 of the adjustment to their buyers, while this number is less than 0.5 for sellers with a small market share. The dynamics are similar for the two measures s_{ij} or s_{ijs}.

While the heterogeneous effects are clear at the seller's side, such asymmetric response is again absent when we rely on customers' market shares x_{ij} or x_{ijs}. This is shown in Figure 14. As noticed, there are no large differences in the pass-through depending on whether the buyers have a large or small degree of market share with respect to each seller i.

Figure 14: Intensive Margin by Consumer Market Share



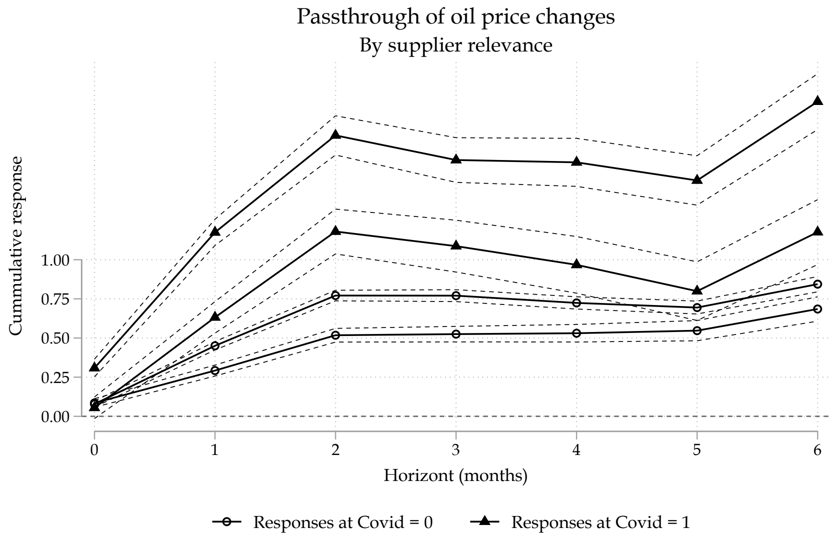


5.2.1 Bilateral market shares and pass-through during COVID-19

Here we show that the role of supplier market share became more important during COVID-19.

Figure 15 shows the results from estimating Equation (5) with three additional controls. One, a dummy variable that takes the value of one during COVID-19 and 0 otherwise. Second, the interaction of that dummy and $\Delta \log \hat{P}^{oil}_{it}$. Third, the interaction of the dummy and $(\Delta \log \hat{P}^{oil}_{it} \cdot Z_{ij})$. We then plot $\frac{\partial \pi^{K,ijv}_{t-1,t+h}}{\partial \Delta \log \hat{P}^{oil}_{it}}(COVID = 0) = \beta^K_h + \gamma^K_h Z_{ij}$, while for the COVID period $\frac{\partial \pi^{K,ijv}_{t-1,t+h}}{\partial \Delta \log \hat{P}^{oil}_{it}}(COVID = 1) = \beta^K_h + \beta^{K,COVID}_h + (\gamma^K_h + \gamma^{K,COVID}_h) \cdot Z_{ij}$.

Figure 15: Supplier market share: COVID-19 vs Non-COVID-19



Although higher bilateral market share is associated with higher pass-through in both periods, during COVID-19, the difference was larger.

6 A simple theoretical framework

Our facts provide evidence of the importance of market shares in driving the frequency of price change and, therefore, the pass-through of shocks.[11]

The baseline model follows Gopinath and Itskhoki (2010) Klenow and Willis (2016), Aruoba et al. (2024). There are two sectors of the economy. The final good sector, composed of a continuum of identical competitive firms, and the intermediate good sector. The intermediate good firms produce differentiated varieties and display heterogeneous levels of productivity and demand. We solve the static version of the model as in

[11]Existing models emphasizing market power and networks often assume flexible prices (Grassi et al., 2017; Alviarez et al., 2023; Dhyne, Kikkawa, and Magerman, 2022, e.g.,). The new generation of network models with sticky prices (e.g., Ghassibe, 2021) rely on monopolistically competitive firms with constant elasticity of demand, which rules out within industry market concentration as a driver of price settings. Mongey (2021) proposes a model with oligopolist firms and endogenous frequency price change as a function of market power. However, such a model abstracts from market power within firm-to-firm relationships. Our facts point to incorporating state-dependent price rigidity in menu cost models of networks and non-constant demand elasticity.

Gopinath and Itskhoki (2010). We interpret the intermediate good producers as producers of a particular subclass in our data.

6.1 Final good sector

The firms in the final good sector combine all the different subclasses i using a Kimball (1995) aggregator:

$$\int_0^1 G\left(\frac{n_t^i y_y^i}{Y_t}\right) d_i = 1. \tag{6}$$

The term n_t^i is a subclass specific demand shock and $\frac{y_y^i}{Y_t}$ is the relative output demanded. We follow Arouba et al and Dotsey and King (2005) and assume

$$G\left(\frac{n_t^i y_y^i}{Y_t}\right) = \frac{\omega}{1+\omega\psi}\left((1+\psi)\frac{n_t^i y_y^i}{Y_t} - \psi\right)^{\frac{1+\omega\psi}{\omega(1+\psi)}} \tag{7}$$

where the elasticity of substitution between subclasses is $\frac{\omega}{\omega-1}$. The final-good maximization problem yields the following relative demand for subclass i

$$\frac{n_i y_i}{Y} = \frac{1}{1+\psi}\left(\left(\frac{p_i}{\lambda n_i P}\right)^{\frac{\omega}{1-\omega}(1+\psi)} + \psi\right), \tag{8}$$

6.2 Intermediate producers' price setting

The intermediate good producers use labor and an imported input (oil) in their production. As in Gopinath and Itskhoki (2010), the marginal cost is

$$MC_i = (1-a_i)(1+\phi e)c$$

is the marginal cost of the supplier i. Besides the productivity shock and the demand shock, the firms are subject to import price shocks e. The profit function of the intermediate input supplier of subclass i is

$$\pi(p_i, a_i, n_i, e, P) = \varphi(\cdot)\frac{1}{P}\Big(p_i - (1-a)(1+\phi e)c\Big), \tag{9}$$

where $\varphi(\cdot)$ is the relative demand from the downstream sector.

For a given cost shock (a, e), the desired price of the firm is

$$\pi(p_i, a_i, n_i, e, P) = arg\max_{p_i} \pi(p_i, a_i, n_i, e, P).$$

Now, define as $\overline{p}_i^0$ as the price the firm sets prior observing the cost shock. If the firm chooses not to adjust the price it will earn $\pi(\overline{p}_i^0 | a_i, n_i, e, P)$. The decision to adjust the price is optimal if the profit gain from adjusting is larger than the menu cost κ:

$$L(a, e) = \pi(p_i, a_i, n_i, e, P) - \pi(\overline{p}_i^0 | a_i, n_i, e, P) > \kappa$$

6.3 Pricing decision

For a given realization of the states, the desired price of the firm is

$$\pi(p_i, z_i, n_i, S) = arg\max_{p_i} \pi(p_i, z_i, n_i, S).$$

The optimal price of the firm is

$$p_i = \frac{\tilde{\sigma}(s_i(p_i))}{\tilde{\sigma}(s_i(p_i)) - 1} mc_i \tag{10}$$

where $\tilde{\sigma}(s_i(p_i))$ is the effective price elasticity of demand and $mc_i = \frac{W^{1-\phi}e^{\phi}}{z_i}$ is the marginal cost of producer i. With Kimball demand structure, the effective demand elasticity is a function of the price p_i and the market share of the firm s_i.

6.3.1 Frequency of price adjustment

We study the frequency of price adjustment in a static set up as in Gopinath and Itskhoki (2010). The decision to adjust the price is optimal if the profit gain from adjusting is larger than the menu cost κ:

$$L(\delta) = \pi(p_i, \delta, n_i, P) - \pi(\overline{p}_i^0|\delta, n_i, P) > \kappa$$

Now, define as $\overline{p}_i^0$ as the price the firm sets prior observing the cost shock. If the firm chooses not to adjust the price it will earn $\pi(\overline{p}_i^0|\delta, n_i, P)$.

The loss function evaluated at p_δ becomes

$$L(\delta|\overline{p}_0) = -\frac{1}{2}\frac{\partial^2 \pi(p|\delta)}{\partial p^2}(p_\delta - \overline{p}_0)^2 + O(p_\delta - \overline{p}_0)^3,$$

$$L(\delta|\overline{p}_0) = \frac{1}{2}\frac{y_i \tilde{\sigma}(s_i)}{p_\delta}\left[\tilde{\varepsilon}(s_i) + \frac{(1+\delta)c}{p_\delta}\left(1 - \tilde{\varepsilon}(s_i)\right)\right](p_\delta - \overline{p}_0)^2 + O(p_\delta - \overline{p}_0)^3,$$

Using the equation for the optimal price in a flexible price world, we have

$$p_\delta = \frac{\tilde{\sigma}(s_i)}{\tilde{\sigma}(s_i) - 1}(1+\delta)c.$$

$$L(\delta|\overline{p}_0) = \frac{1}{2}\frac{s_i Y(\tilde{\sigma}(s_i) - 1)}{(1+\delta)cn_i}\left[\tilde{\varepsilon}(s_i) + \frac{(\tilde{\sigma}(s_i) - 1)\left(1 - \tilde{\varepsilon}(s_i)\right)}{\tilde{\sigma}(s_i)}\right]\underbrace{(p_\delta - \overline{p}_0)^2}_{Passthrough},$$

where $\tilde{\varepsilon}(s_i) = \frac{\omega}{1-\omega}\frac{\psi}{ny/Y}$, $\tilde{\sigma}(s_i) = -\frac{\partial y_i}{\partial p_i}\frac{p_i}{y_i} = \frac{\omega}{\omega-1}\frac{(1+\psi)\frac{ny_i}{Y} - \psi}{\frac{n_i y_i}{Y}}$, and the pass through is from Proposition 1 in Gopinath and Itskhoki (2010), first order approximation:

$$p_\delta - \overline{p}_0 = \frac{1}{1 + \frac{\tilde{\varepsilon}(s_i)}{\tilde{\sigma}(s_i) - 1}}(1+\delta)$$

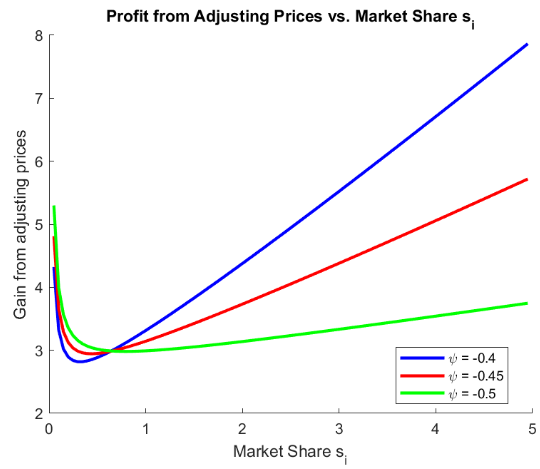


Figure 16: Loss of not adjusting prices

The frequency of price adjustment is then defined as

$$\Phi = 1 - Pr\{\delta \in \Delta\} = Pr\{L(\delta|\overline{p}_0) > \kappa\}$$

Figure 16 plots the relationship between the loss function L and the market share of supplier i. Note that in a symmetric equilibrium, when every supplier has the same importance in the market, $s_i = \frac{n_i y_i}{Y} = 1$. A s_i of 10, for example, indicates that a firm is 10 times larger than the "average firm" or the steady state firm. Here we study the partial equilibrium relationship without taking a stand into what drives s_i and what are the GE effects of the cost shock δ.

Figure **??** depicts the two different components of the loss function: the curvature of the profit function, first large term in bracket in L and the passthrough.

Note that these are parameter combinations that fit our facts but they are not far from those used in Edmond et al (2023) JPE. We do need a ψ that is larger than in Aruoba et al. (2021).

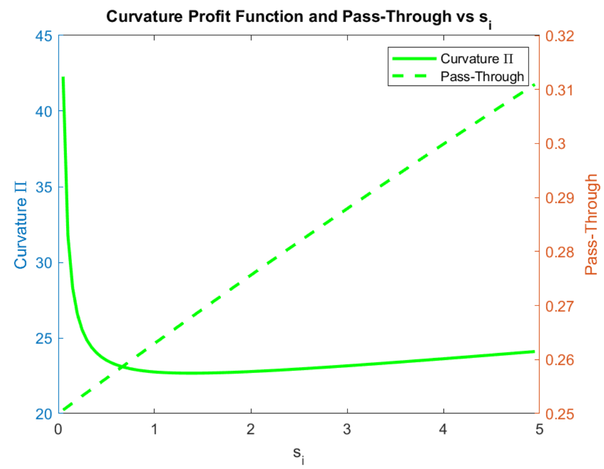


Figure 17: Market share and price

6.4 One supplier and many buyers

Here we make the model more comparable to the evidence we present. Now, the intermediate good producers set their prices to buyer 1 and buyer 2 and faces menu costs ($\kappa_{i1} > 0$ and $\kappa_{i2} > 0$) every time they decide to adjust their prices. The production function of a intermediate good producer is

$$y_i = z_i \left(\frac{l_i}{1-\phi}\right)^{1-\phi} \left(\frac{m}{\phi}\right)^{\phi}.$$

The cost of labor is W and the imported input price is e. The profit function of the intermediate input supplier of subclass i is

$$\pi(p_i, z_i, n_i, S) = y_{i1}\frac{1}{P}\left(p_{i1} - \frac{W^{1-\phi}e^{\phi}}{z_i}\right) + y_{i2}\frac{1}{P}\left(p_{i2} - \frac{W^{1-\phi}e^{\phi}}{z_i}\right),$$

where S represents the aggregate space (W, P, e). The demand from the downstream buyer j for the particular subclass i is

$$s_{ij} = \frac{n_{ij}y_i}{Y_j} = \frac{1}{1+\psi_j}\left(\left(\frac{p_{ij}}{\lambda_j n_{ij} P}\right)^{\frac{\omega_j}{1-\omega_j}(1+\psi_j)} + \psi_j\right), \qquad (11)$$

$$p_{ij} = \frac{\tilde{\sigma}(s_{ij}(p_{ij}))}{\tilde{\sigma}(s_{ij}(p_{ij})) - 1} mc_i \tag{12}$$

where $\tilde{\sigma}(s_{ij}(p_{ij}))$ is the effective price elasticity of demand and $mc_i = \frac{W^{1-\phi}e^{\phi}}{z_i}$ is the marginal cost of producer i. With Kimbal demand structure, the effective demand elasticity is a function of the price p_{ij} and the market share of the firm s_{ij}.

We study the frequency of price adjustment. For a particular relationship we have

$$L_{ij}(\delta) = \pi(p_i j_{j=1}^{J}, \delta, n_{ij}, P) - \pi(\overline{p}_{ij}^{0} | \delta, n_{ij}, P) > \kappa_{ij}.$$

Say $L_{i1}(\delta) > \kappa_{i1}$ but $L_{i2}(\delta) < \kappa_{i2}$ such that $L_{i1} + L_{i2} > \kappa_{i1} + \kappa_{i2}$. It is optimal for the supplier to only adjusting p_{i1} instead of adjusting prices to both suppliers.[12]

7 Conclusion

Our findings emphasize the critical role of firm-to-firm relationships in shaping price adjustment behavior, highlighting the importance of match-specific characteristics in driving the heterogeneity of price adjustment decisions. Specifically, we show that suppliers' market share in their clients' purchases is a key determinant of the likelihood of price adjustments in response to external shocks, such as oil price fluctuations. The asymmetric pass-through of positive versus negative shocks further emphasizes the influence of bilateral market power on price-setting dynamics. These results suggest that traditional models of state-dependent pricing, which incorporate menu costs and variable demand elasticities, should consider the role of market power within supplier-client networks. The implications for understanding inflation dynamics and the transmission of monetary policy can be significant, particularly given the pronounced heterogeneity in firm-to-firm relationships compared to sectoral-level differences. This is especially relevant during pe-

[12]This result would break if one incorporates one menu cost κ of adjusting one or many prices of supplier i, as in the multiproduct literature. This is an object of future research.

riods of substantial shocks, such as the COVID-19 pandemic, where the granular nature of these relationships seems to play a more critical role in shaping economic outcomes.

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Table 7: Extensive Margin Variance Decomposition: Subclass level

Table 8: Extensive Margin Variance Decomposition

Appendix

Figure 18: Size and subclasses

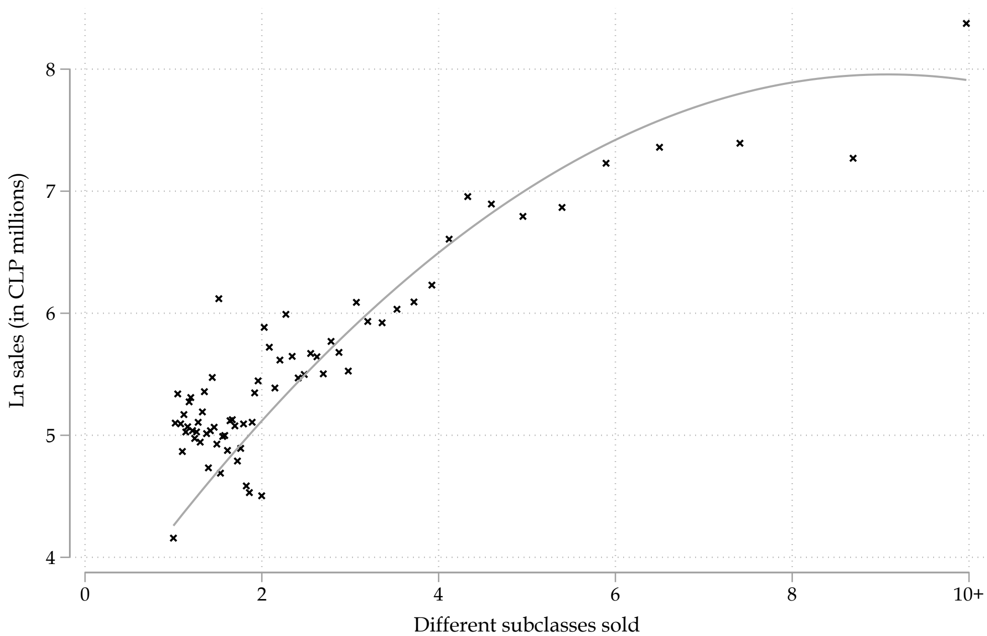
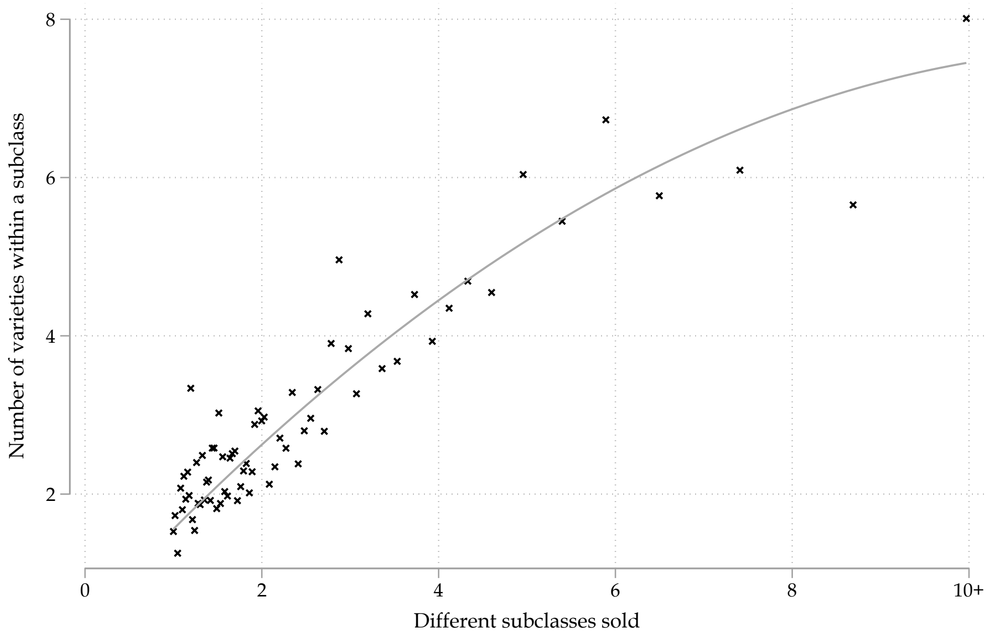


Figure 19: Variety and subclasses



7.1 Competitors price gap

Here we study the extent of strategic complementarities in our data. Figure 20 shows the distribution of competitors' price gap, as defined in Karadi, Schoenle, and Wursten (2024). For each firm, we measure the average price of the competitors within the product subclass market. As seen in Figure 20, we observe that the price gap is centered around zero, with most of the prices below or above 50% the average price of the market.

In Figures 21 and 22 we confirm the results in Karadi, Schoenle, and Wursten (2024). The magnitude of the price change increases with the distance from the average price of the competition. Also, firms are more likely to change the price when they deviate from the competitor's price.

As Karadi, Schoenle, and Wursten (2024) emphasize, these facts are consistent with state-dependent price-setting models in which menu costs are random (e.g., Dotsey, King, and Wolman, 1999; Luo and Villar, 2021; Alvarez, Lippi, and Oskolkov, 2022).

Figure 20: Competitor-price gap (strategic complementarities)

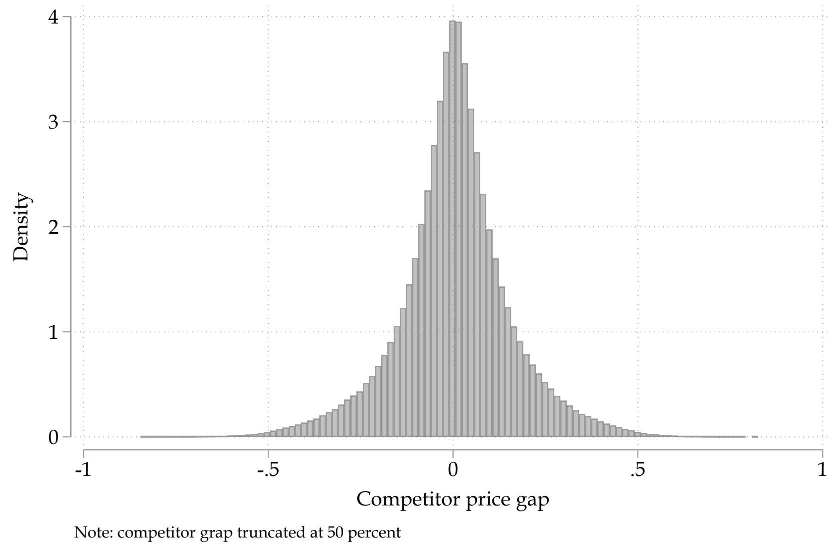
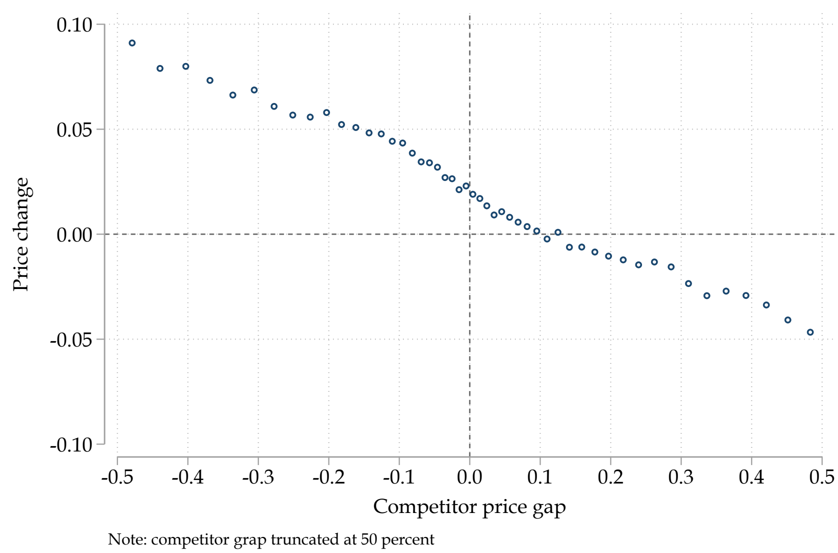


Figure 21: Magnitude of adjustment (t+1) as function of the gap

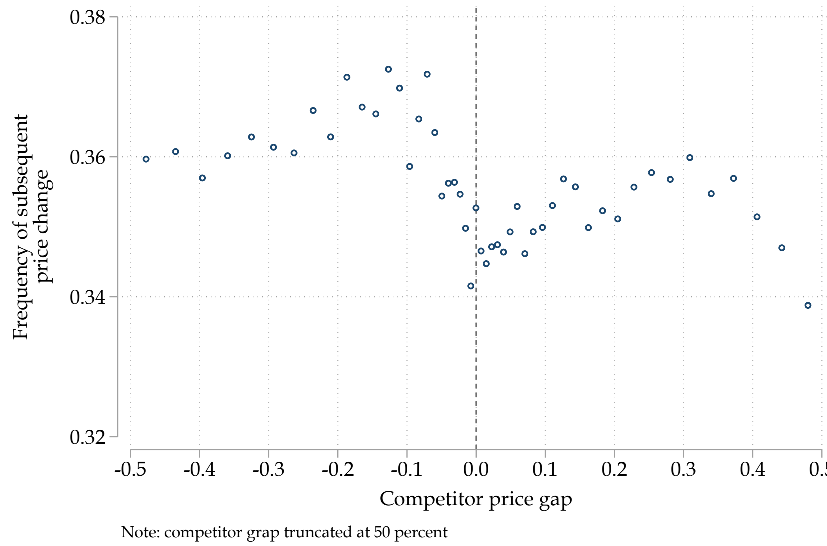


Version: Filtered of supplier-product FE using OLS

7.2 Sectoral heterogeneity

We now explore the heterogeneity in sectoral frequency price change within the PPI sample. Figure 23 reports the median and the weighted average frequency price change for the 111 sectors in our sample. Two facts stand out. First, as documented by Pas-

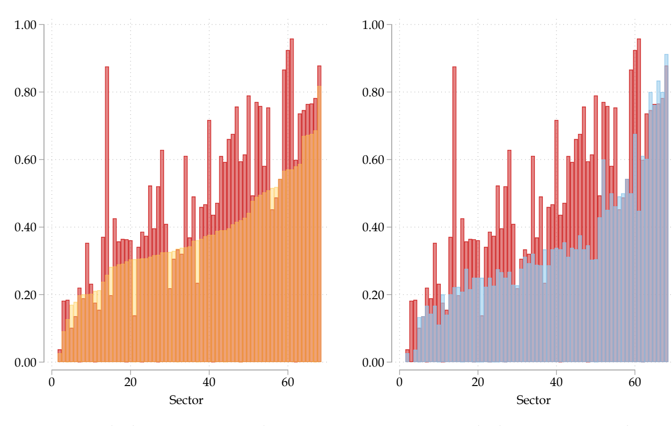
Figure 22: Magnitude of adjustment as function of the gap



Version: Filtered of supplier-product FE using OLS

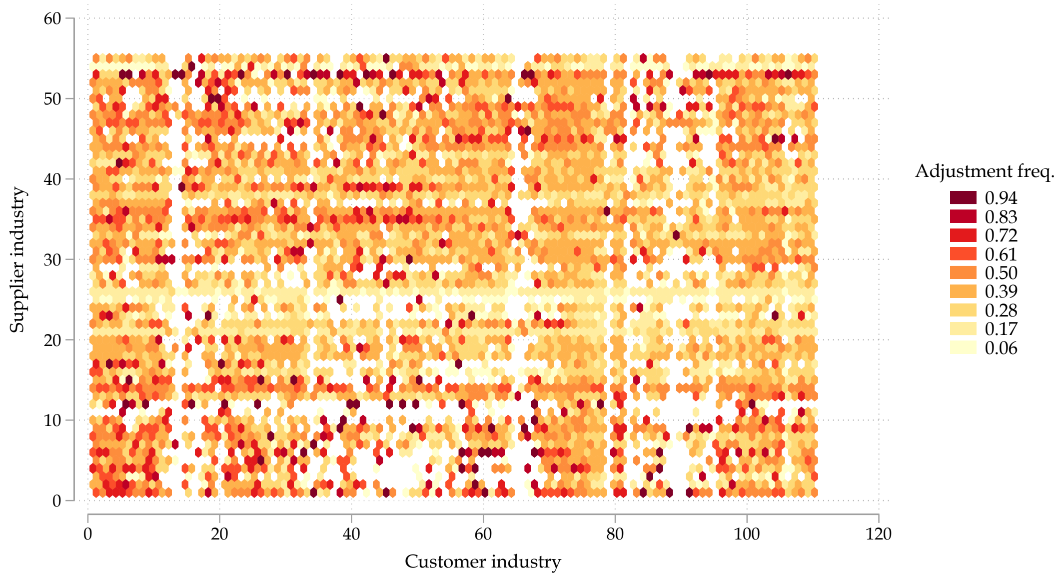
ten, Schoenle, and Weber (2020), there is large sectoral heterogeneity in frequency price changes. Second, there is still significant heterogeneity within sectors, as expressed by the difference between the median and the weighted average frequency of price change. Most sectors display a weighted average that is significantly larger than the median. For example, in many sectors, the weighted average is up to 3-4 times larger, indicating that large firms tend to change prices more frequently within sectors.

Figure 23: Frequency of adjustment between sectors (CAE 111 industries)



We now document an additional degree of heterogeneity in sectoral frequency price changes. We calculate the frequency price change at the i supplier-client level. Figure 24 plots the sector-to-sector frequency price change. The vertical axis represents the supplying sector and the horizontal axis represents the customer sector. Light colors indicate low-frequency price change and darker colors represent high-frequency price change. We observe significant heterogeneity in frequency price changes from a given supplier sector to different customer sectors.

Figure 24: Sector-to-sector frequency of adjustment between sectors (CAE 111 industries)



f_{IJ}: monthly frequency of price IJ changes ($> 0.5\%$) for all IJ transactions (average across goods and time: 2018.m1-2023.m6)

7.3 Stylized facts - Conditional Price Adjustments

7.4 Stylized facts about s_{ijvt} and x_{ijvt}

In this Section we provide stylized facts for two measures of bilateral market share at the seller level s_{ijvy} and at the buyer level x_{ijvy}.

Table ?? provides descriptive statistics of the bilateral market shares. Clearly, the seller's bilateral market share is more highly skewed than the buyer's counterpart. While

Table 9: Sample Descriptive Stats

Stat	Intensive	Extensive
Transactions (thousands)	27.6	287.3
Sellers	126	287
Buyers	1,686	11,197
Varieties	441	2.289
Buyer-seller pairs	1,717	12,129
Buyer-variety pairs	2,223	25,162
Subclass	27	119
Varieties per subclass	16.3	19.3

the mean of s_{ijvy} is 0.16 in the sample, the mean of x_{ijvy} is approximately 0.01 for the same period. At higher percentiles of the distribution of s_{ijvy}, we found a share bigger than half, which turns to one for the 95th percentile. As expected, given a buyer, there's a non-negligible amount of products constantly purchased by the same buyer.

Table 10: $IJVT$ Market Share Stats

Var	Mean	StdDev	p05	p10	p25	p50	p75	p90	p95	N
s_{ijvy}	0.155	0.278	0.001	0.002	0.007	0.029	0.137	0.582	1	30,833,286
x_{ijvy}	0.008	0.069	0.000	0.000	0.000	0.000	0.000	0.002	0.009	30,833,286

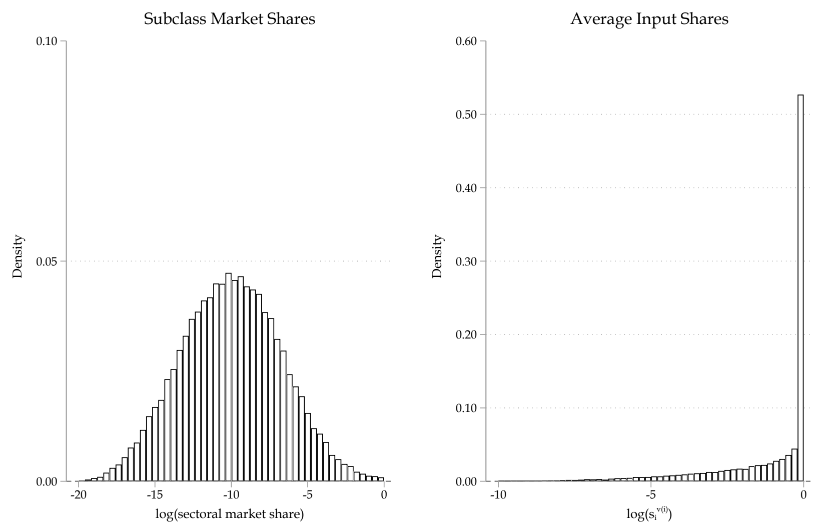
The latter is not true for buyers. The buyers' market share, in terms of quantities, displays an average of 0.008 and a median of 0.0001. Only at the top 10% and 5% of buyers' market shares, we observe a bilateral market share of 0.002 and 0.009, respectively. Therefore, unlike the supplier market share, most buyers are atomistic, where most buyers demand most of their input varieties within a subclass of products from different sellers instead of concentrating on only one.

UPDATE Figure 25 depicts in panel a) the distribution of firm-level market share at the subclass level.[13] In panel b), we plot the distribution of a given supplier's average bilateral market share with all his clients s_{ijv}. There are clear differences between the distribution of total market shares and the supplier-client market shares. There is signif-

[13] This is simply the total sales of seller i of subclass s over the total amount of sales of product s in the economy.

icantly more variation in the former, while the latter is characterized by a large mass of suppliers holding a large share of clients' purchases. This result is consistent with that documented in Dhyne, Kikkawa, and Magerman (2022), except that our finer level of disaggregation uncovers even larger differences between the two measures.

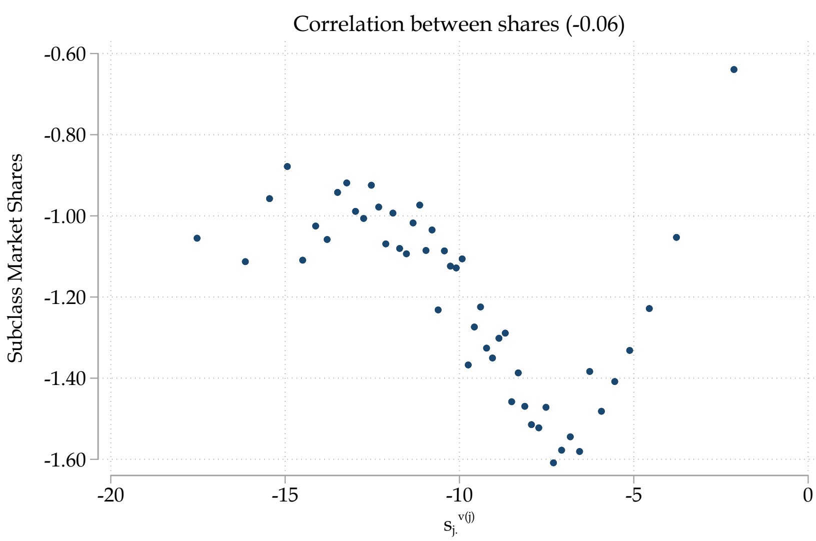
Figure 25: Subclass share comparison



Note: This figure plots the histogram of the logarithm of firms' product market shares, panel a), and the logarithm of the average product market share within supplier-client relationships, panel b). Both measures are averaged over time to obtain one measure per supplier or per supplier-client pair.

In Figure 26 we show the U-shaped relationship between the two measures of market shares. The overall correlation is -0.06. For a significant part of the distribution of firm-level size, we observe that being a large firm in the market is associated with smaller market power at the supplier-client level. For suppliers with a very large average bilateral market share we do observe a positive relationship between the overall market share and the average bilateral market share.

Figure 26: Subclass share correlation



Note: This figure plots the binscatter plot between product market shares and the average product market share within supplier-client relationships.